

Automated quality evaluation of dental panoramic radiographs using deep learning

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ABSTRACT

Purpose: Panoramic radiographs are instrumental in dental diagnosis but face quality issues related to contrast, artifacts, positioning, and coverage, which can impact diagnostic accuracy. Although expert assessment is the accepted standard, it is time-consuming and prone to inconsistency. Artificial intelligence offers an automated, objective solution for evaluating radiograph quality, increasing efficiency and reducing inter-rater variability.

Materials and Methods: This study aimed to develop a deep learning (DL)-based model for evaluating the quality of dental panoramic radiographs. A dataset of 1,000 panoramic images, collected from 2018 to 2023, was assessed by 2 trained dentists using predefined grading criteria for contrast/density, artifact presence, coverage area, patient positioning, and overall quality. These expert-annotated scores were used as the ground truth to train and validate 5 YOLOv8 classification models, each targeting a specific quality criterion. The models' performance was evaluated on a separate test set using performance metrics.

Results: The YOLOv8 models achieved classification accuracies of 87.2%, 74.1%, 77.3%, 97.9%, and 79.3% for artifact detection, coverage area, patient positioning, contrast/density, and overall image quality, respectively. The model used to classify images as clinically acceptable or unacceptable exhibited an average accuracy of 81.4%, demonstrating its potential for real-world application.

Conclusion: These findings highlight the feasibility of DL-based automated image quality assessment for panoramic radiographs. The high accuracy of the proposed model suggests its potential integration into clinical workflows to assist practitioners in efficiently evaluating radiograph quality. Additionally, such a model could represent an educational tool for dental students, improving radiographic techniques and reducing unnecessary retakes. (*Imaging Sci Dent* 2025; 55: 175-88)

KEY WORDS: Image Processing; Radiography, Panoramic; Deep Learning; Artificial Intelligence

Introduction

Panoramic radiography is a conventional dental imaging modality that captures a single image of the maxillomandibular region. Due to its capacity to provide a comprehensive view of the teeth, alveolar bone, temporomandibular joints, and surrounding structures, it is the most commonly used extraoral radiographic modality in dentistry.¹ The in-

terpretation of panoramic radiographs can be challenging due to the complex relationships among anatomical landmarks and the superimposition of structures, necessitating extensive training and experience.^{2,3} Accurately interpreting radiographs is a critical aspect of medical diagnosis and treatment planning, with the quality of the images playing a key role. Inadequate radiographic quality can lead to issues including misinterpretation, misdiagnosis, and poor treatment planning.⁴

The development of digital radiography has simplified the capture of 2-dimensional (2D) images and accelerated their processing. This modality provides dental professionals with tools to enhance radiograph quality, improving the accuracy of diagnostic interpretation. Additionally, digital radiography has made it easier to store, manage, and trans-

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mit images electronically.⁵⁻⁷ Consequently, the number of digital extraoral radiographs taken has risen significantly in recent years.⁸ This shift to digital platforms also offers a valuable educational resource, allowing dental students to access and review a wide array of radiographic images and thereby augment their learning and interpretive skills.⁹

The availability of a large volume of images enables the development of machine learning algorithms and artificial intelligence (AI) systems that can assist in clinical decision-making. These systems can analyze radiographs, identify potential abnormalities, and provide recommendations for treatment or further diagnostic testing.¹⁰ The application of deep neural networks to medical data, particularly through convolutional neural networks (CNNs), is increasing for tasks such as image classification, segmentation, detection, and anomaly identification.¹⁰⁻¹³ Among these models, the You Only Look Once (YOLO) series consists of real-time single-stage object detection algorithms that have gained popularity in the medical field for their speed and accuracy.¹⁴ Studies have shown that these algorithms can be employed for various tasks in dentistry, including tooth detection and numbering,^{10,11} diagnosing dental caries,¹⁵ assessing root morphology,¹⁶ detecting and classifying radiolucent lesions,^{17,18} identifying the inferior alveolar nerve and third molar,¹⁹ and detecting periodontal bone loss.²⁰ Incorporating these advanced AI tools into dental education can also help students familiarize themselves with cutting-edge technologies, preparing them for future advancements in dental diagnostics and treatment planning.

The accuracy and reliability of these models depend on the use of high-quality, homogeneous images for training and validation,¹² making quality control of input images a crucial step before using radiographs for model training.²¹ Image quality assessment (IQA) can be divided into objective and subjective approaches. In subjective assessment, human experts evaluate the quality of a radiograph to determine its clinical acceptability. With objective quality assessment, a model is developed to simulate this evaluation process by learning features from a labeled training dataset and using them to map scores to new data.²² The subjective approach, which involves manual assessment, has been used to evaluate the quality of 2D dental radiographs.²³⁻²⁵ However, this method is time-consuming, especially given the large number of images required for training, and prone to error. While the performance of objective IQA tools in medical imaging has been investigated in several studies,²⁶⁻³⁰ research in dentistry remains limited. To address this gap, the present study aims to develop and validate a

deep learning (DL)-based model for the automated quality assessment of panoramic radiographs. Specifically, the authors sought to classify images based on key quality criteria, including artifacts, coverage area, patient positioning, and contrast/density, while also assessing overall image acceptability for clinical diagnosis. By leveraging a YOLOv8 model, the goal is to enhance IQA for both clinical and educational settings, reducing diagnostic errors and improving workflow efficiency.

Materials and Methods

In a secondary data study approved by the University Ethics Board, 1,000 dental panoramic images, captured over the past 5 years (2018-2023), were extracted from the University of Alberta Mike Petryk School of Dentistry Oral Health clinic database. The digital panoramic radiographs were acquired using 3 OrthoPhos XG Series (Dentsply Sirona, Bensheim, Germany) devices. The radiographs were obtained by calibrated and experienced technicians in accordance with the manufacturer's recommended settings.

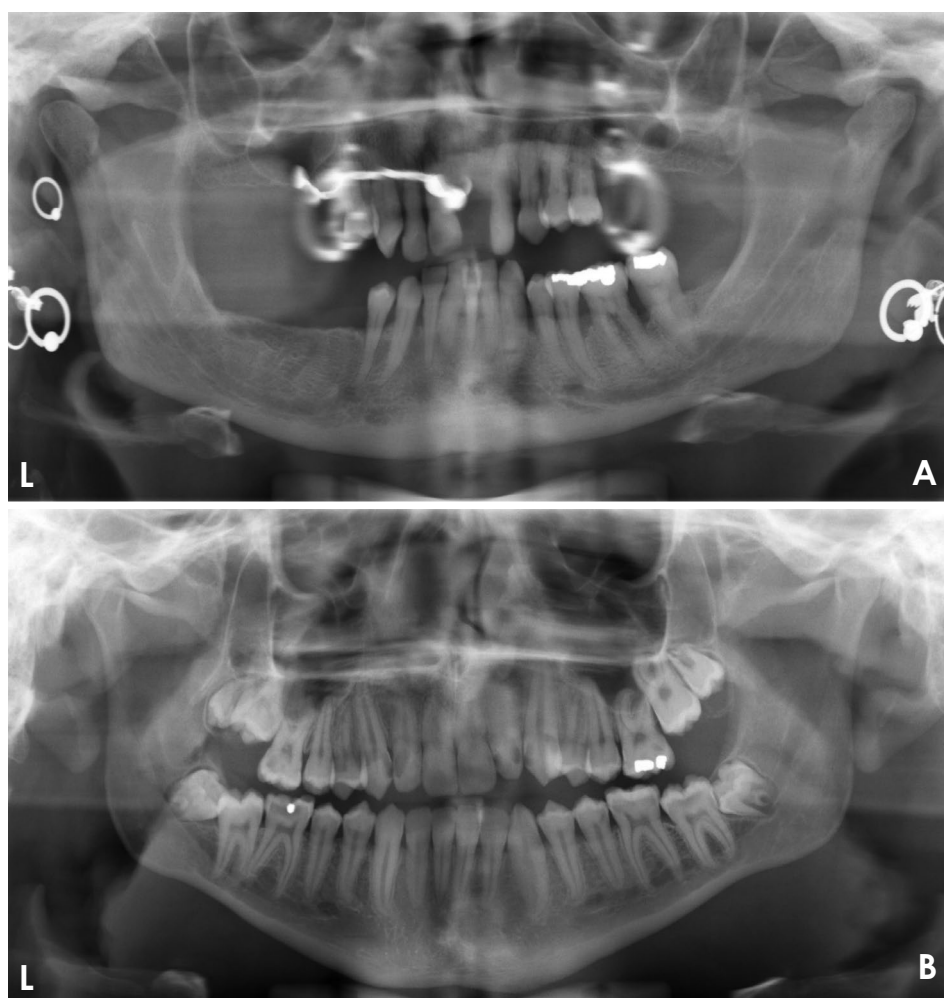
The present study employs 4 parameters—coverage, artifacts, patient positioning, and contrast/density—to classify images. An overall IQA was also conducted to determine clinical acceptability for diagnostic purposes.

As IQA relies on dental structures, radiographs that lacked teeth (as in edentulous patients) were excluded from the available pool. Radiographs were also excluded if they contained primary teeth or mixed dentition, since the present study focused on permanent dentition to ensure consistency in evaluation criteria; if they had severe artifacts or distortions that rendered quality assessment infeasible; or if they were duplicated or improperly labeled, to prevent bias and redundancy in model training. Following calibration sessions, the quality of each image was independently assessed and scored by 2 dentists (N.A. and M.M.), both trained by an expert (C. P.-P.) oral and maxillofacial radiologist (OMFR), using the predefined set of criteria outlined in Table 1. The criteria were developed based on the research of Choi et al.³¹

Regarding patient positioning, radiographs were evaluated for chin positioning errors (too high or too low), which lead to excessive curvature or flattening of the occlusal plane; midline positioning errors, which produce asymmetrical distortion of anatomical structures; patient rotation or tilt, which causes one side of the mandible to appear magnified while the other appears compressed; and anterior-posterior positioning errors (too far forward or backward), which affect the magnification and sharpness of an-

Table 1. Parameters for the quantitative quality assessment of panoramic radiographs

Evaluation parameter	Image scores	Description
Coverage area	0, 1, 2	0: Inappropriate coverage and irrelevant to clinical needs 1: Visibility of coverage relevant to clinical needs 2: Appropriate and optimal coverage depending upon the clinical application
Density and contrast	0, 1	0: Poor density and inadequate contrast between enamel and dentin 1: Satisfactory density and adequate contrast between enamel and dentin
Patient positioning	0, 1	0: Inappropriate patient position 1: Symmetric, no changes in the occlusal plane or chin position, no patient movement (continuity of borders/no dual images)
Overall image quality	0, 1, 2, 3, 4, 5	≤2: Clinically unacceptable, too poor for diagnosis (image retake required) >2: Clinically acceptable (no image retake required)

**Fig. 1.** Examples of radiographs demonstrating the presence (A) and absence (B) of artifacts.

terior teeth.⁴

In instances of discordance between the 2 evaluators' assessments, the OMFR re-evaluated the images and established a final score. These quality scores served as the ground truth for validating the classification model. The

image quality scores were computed by aggregating individual scores regarding the 4 criteria (artifacts, coverage area, patient positioning, and contrast/density). Subsequently, images were categorized into 2 groups—clinically acceptable or not acceptable—based on the necessity of

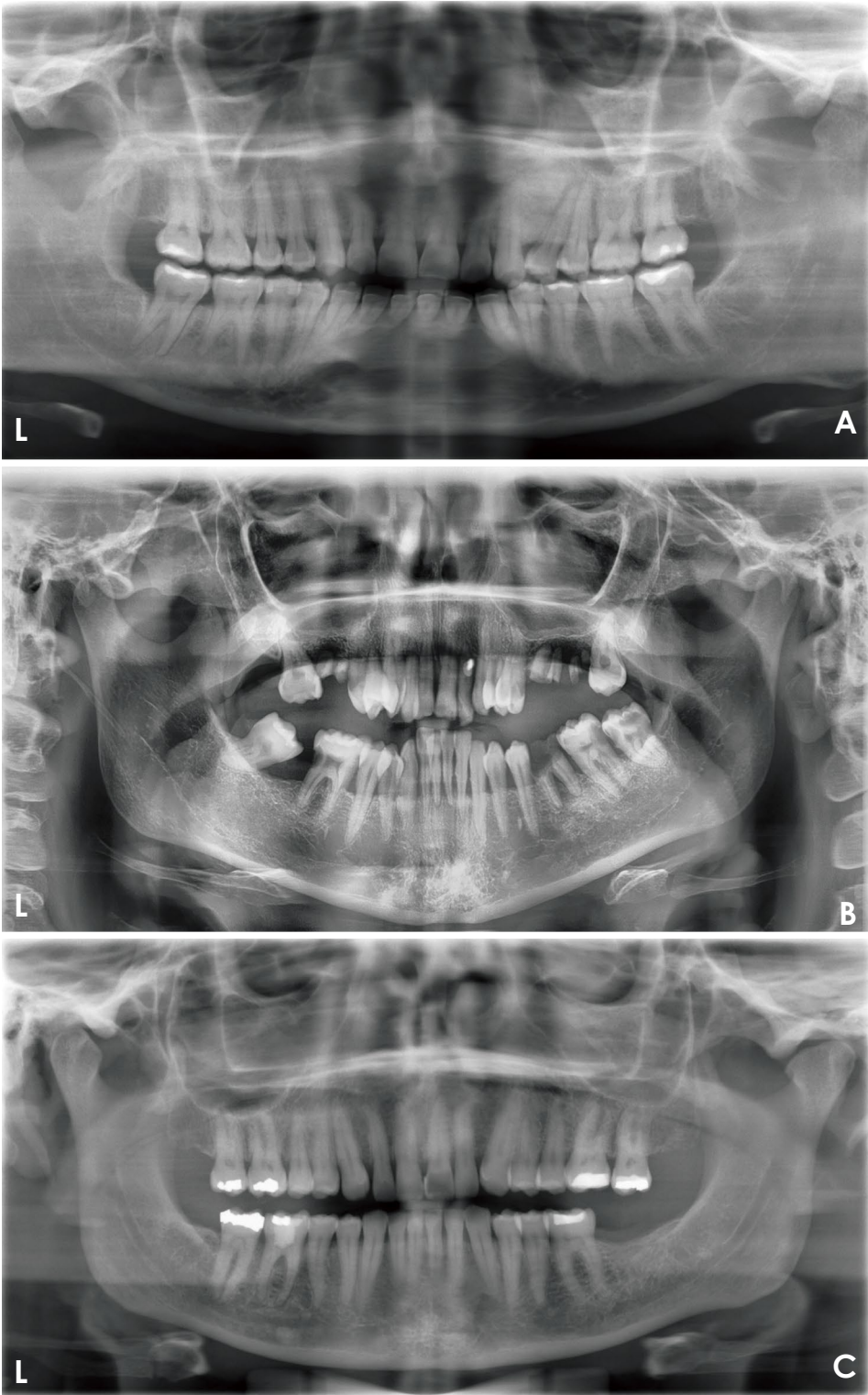


Fig. 2. Examples of radiographs demonstrating poor (A), moderate (B), and good (C) coverage area.

retaking the image. To preserve the original image quality, no preprocessing steps were implemented other than image resizing and augmentation. This limited preprocessing was chosen to maintain the integrity of radiographic features essential for quality assessment. Applying contrast enhance-

ment, noise reduction, or other image manipulations could introduce artificial changes that do not reflect the true quality of the original radiographs. Essentially, this approach ensures that the model learns to classify images based on real-world variations rather than on artificially enhanced

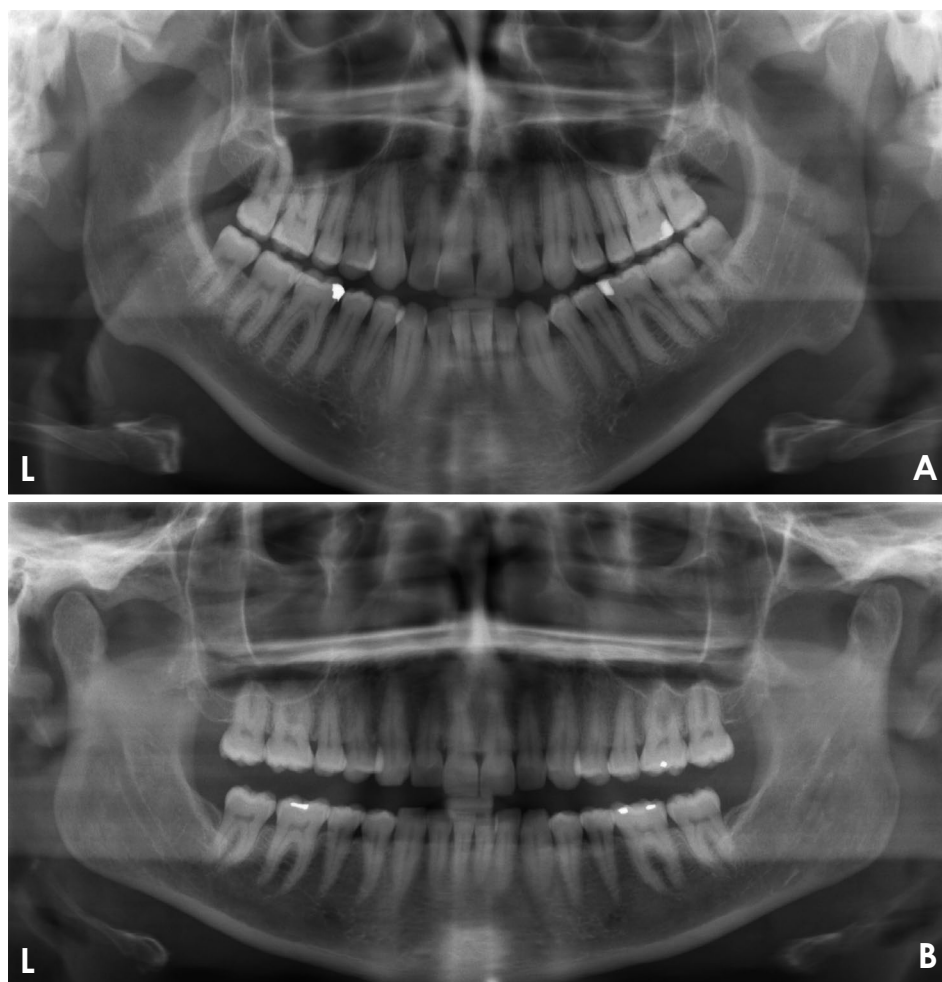


Fig. 3. Examples of radiographs demonstrating poor (A) and good (B) patient positioning.

representations. The 4 criteria for the scoring and classification of radiographs are detailed below.

Artifacts

The raters classified the panoramic images into 2 groups (class 0, artifact present; class 1, no artifacts). Among the 1,000 images, 139 were found to have artifacts (class 0). To balance the classes, 140 images without artifacts were randomly selected. The images were randomly assigned to training (70%, representing 196 images), validation (20%, or 56 images), and test (10%, or 27 images) datasets. Image augmentation was performed after data splitting to ensure that no data leakage occurred. Ultimately, a total of 1,580 images were obtained to train a YOLOv8 classification model.

The model's performance was assessed on a set of 40 new images (20 images from each class) using evaluation metrics. Fig. 1 presents examples of radiographs with and without artifacts.

Coverage area

The raters classified the panoramic images into 3 groups (with classes 0, 1, and 2 indicating poor, moderate, and good coverage areas, respectively). Among the 1,000 images, 99 and 170 images were found to have poor (class 0) and moderate (class 1) coverage areas, respectively. To achieve a balanced distribution across classes, 140 images with good coverage areas were also included. The images were randomly assigned to training (70%, representing 219 images), validation (20%, or 63 images), and test (10%, or 38 images) datasets. Again, after data splitting, image augmentation was performed to prevent data leakage. Ultimately, a YOLOv8 classification model was trained on a dataset comprising 1,525 training images.

To assess the performance of the model on a new test dataset, 31 new images (9 from class 0, 10 from class 1, and 12 from class 2) were used, and the outcome was reported using evaluation metrics.

Fig. 2 presents examples of radiographs with poor, moderate, and good coverage areas as scored by the experts.

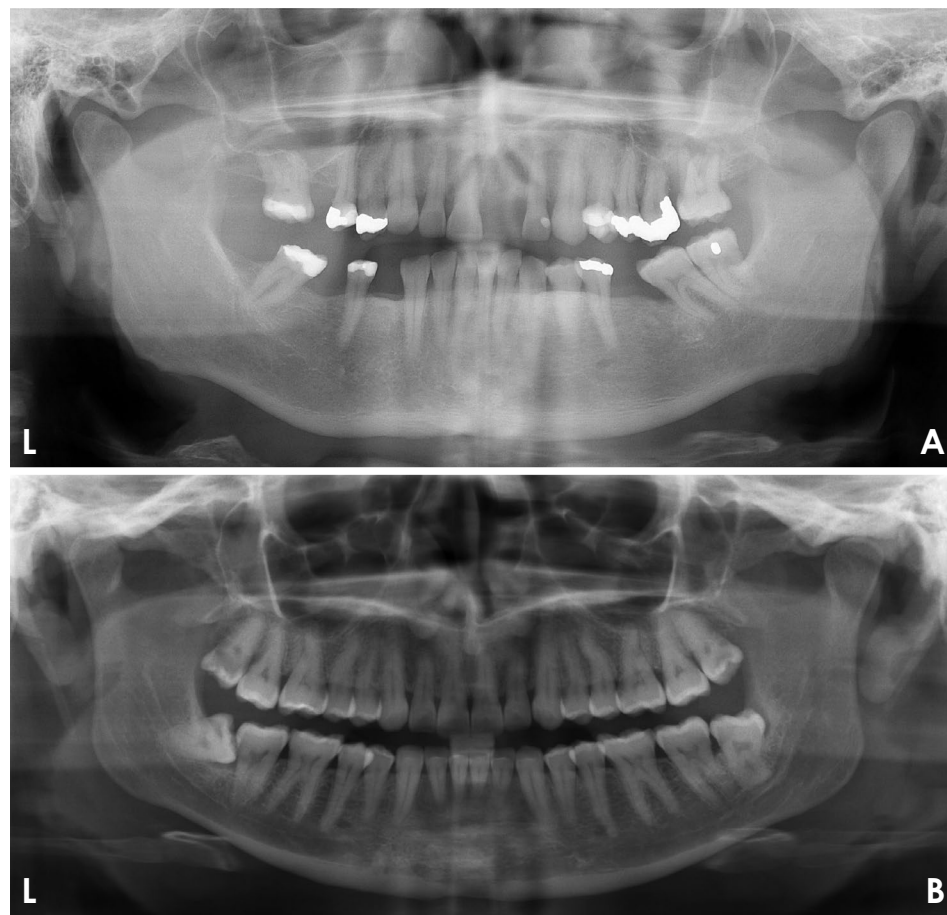


Fig. 4. Examples of radiographs demonstrating poor (A) and good (B) contrast/density.

Patient positioning

To classify the images according to patient positioning, the raters categorized the panoramic images into 2 groups (class 0, poor positioning, class 1, good positioning). A total of 191 images were found to have poor patient positioning (class 0); accordingly, 200 random images with good positioning were also selected. The images were split into training (70%, representing 274 images), validation (20%, or 78 images), and test (10%, or 39 images) datasets. Image augmentation was applied after splitting, resulting in a final dataset of 1,450 images for training the YOLOv8 classification model.

The model's performance was assessed on a set of 23 new images (11 from class 0 and 12 from class 1) using evaluation metrics. Fig. 3 presents examples of radiographs with poor and good patient positioning.

Contrast and density

The raters classified the panoramic images into 2 groups (class 0, poor contrast/density; class 1, good contrast/density). Since 88 images with poor contrast were identified in the pool of 1,000 images, 100 images with good contrast/

density were also randomly selected. The images were split into training (70%, representing 132 images), validation (20%, or 38 images), and test (10%, or 18 images) datasets. After splitting the dataset and performing image augmentation, the YOLOv8 classification model was fine-tuned on a proprietary dataset over 30 epochs. This dataset comprised 905 training images and 335 validation images across the 2 classes.

The model's performance was evaluated on a set of 22 new images (13 from class 0 and 9 from class 1) using evaluation metrics. Fig. 4 presents examples of radiographs with poor and good contrast/density.

Overall quality of the radiographs

To categorize the images into 2 groups based on overall quality, the scores for the previously described criteria were summed. Radiographs with scores of 0, 1, or 2 were assigned to class 0 (clinically not acceptable), while those with higher scores were assigned to class 1 (clinically acceptable). Classification was based on the necessity of repeating image acquisition for proper diagnosis. Among the 1,000 images, 196 were found to be clinically unacceptable

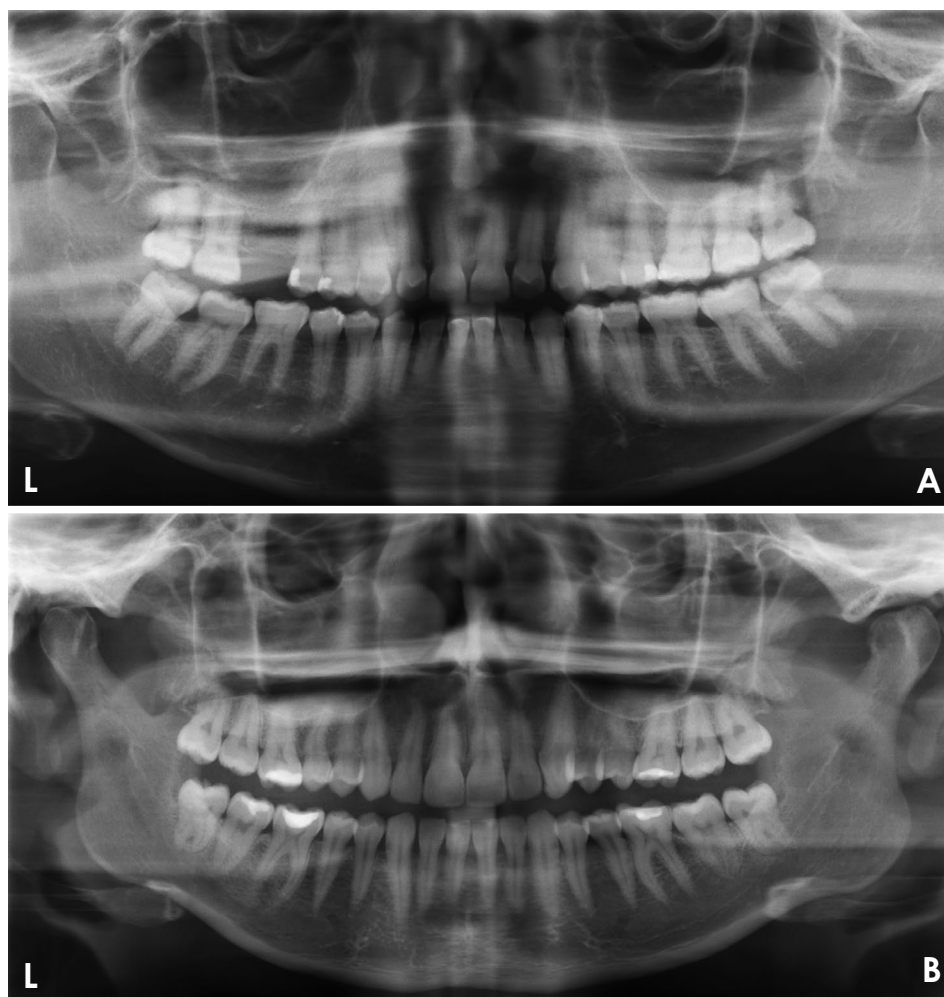


Fig. 5. Examples of clinically acceptable (A) and unacceptable (B) panoramic radiographs.

(class 0); thus, to balance the classes, 200 clinically acceptable images were also included. The dataset was split into training (276 images), validation (79 images), and test (41 images) datasets, and after data augmentation using the approaches described above, a YOLOv8 classification model was trained on a set of 1,641 images.

Finally, the model's performance was evaluated on a set of 96 new images (49 from class 0 and 47 from class 1) using evaluation metrics. Fig. 5 presents examples of clinically acceptable and unacceptable panoramic radiographs.

Model architecture

The model architecture for classifying images according to the abovementioned criteria was a YOLOv8 classification model with 141 layers. The training process involved 30 epochs, a batch size of 16, and the use of an Automatic Mixed Precision approach with the Adam-W optimizer. Hyperparameter optimization was utilized to fine-tune the model for optimal performance. Initially, different learning

rates (ranging from 0.00001 to 0.001) were tested using a cyclic learning rate scheduler; a learning rate of 0.0001 was selected because it led to the lowest validation loss. The batch size was determined after evaluating training stability and GPU memory constraints, with a batch of 16 providing the best balance between computational efficiency and gradient convergence. The Adam-W optimizer was chosen over standard Adam and SGD due to its superior weight decay properties, which help prevent overfitting and increase generalization. To further promote generalization, augmentation techniques including scaling, translation, flipping, and mosaic augmentation were applied. Throughout the training, performance metrics were monitored to ensure good model optimization. The final weights were saved, and the best-performing model was validated on the validation set.

Statistical analysis

The consistency between the 2 raters in scoring the im-

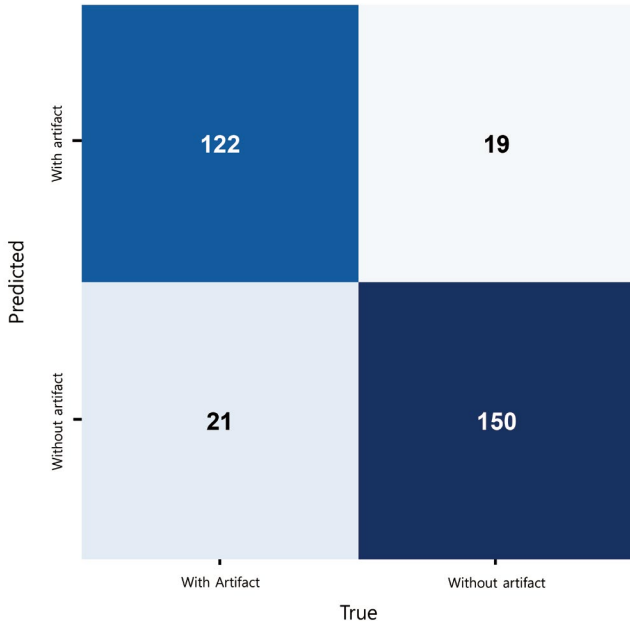


Fig. 6. Confusion matrix for artifact classification on panoramic radiographs.

Table 2. Model performance on classifying panoramic images according to artifact presence

Metrics	Class 0 (with artifact)	Class 1 (without artifact)
Precision	0.792	0.937
Recall	0.950	0.950
F1 score	0.864	0.943
Accuracy	0.889	0.889

ages was assessed using the Cohen kappa, a measure of inter-rater reliability (IRR). This analysis was conducted using a sample of 217 randomly selected images for each of the 4 criteria. The IRR was measured using Python and the sklearn.metrics library.

Classification accuracy measures were used to evaluate outcomes from the validation image set. In machine learning, different evaluation metrics are applied according to the type of problem; for classification tasks, accuracy, precision, recall, and F1 score are commonly used. As this study was based on a classification task, these evaluation criteria were used to assess the performance of all models. A confusion matrix was employed to calculate these metrics.³² This matrix contains true positive (TP), true negative (TN), false positive (FP), and false negative (FN) values. The equations for accuracy, recall, precision, and F1 score are as follows: $\text{accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$; $\text{recall} = \text{TP} / (\text{TP} + \text{FN})$; $\text{precision} = \text{TP} / (\text{TP} + \text{FP})$; and $\text{F1 score} = 2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall})$.

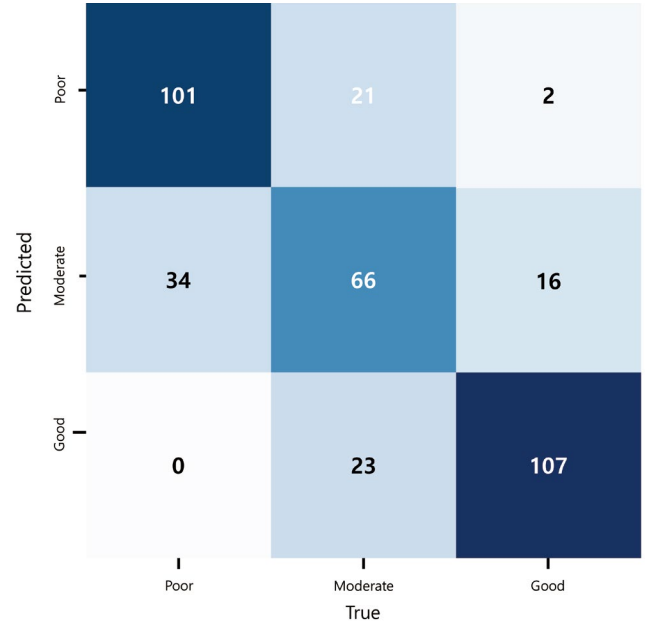


Fig. 7. Confusion matrix for coverage area classification on panoramic radiographs.

Table 3. Model performance on classifying panoramic images according to coverage area

Metrics	Class 0 (poor coverage area)	Class 1 (moderate coverage area)	Class 2 (good coverage area)
Precision	1.000	1.000	0.714
Recall	0.889	0.700	0.833
F1 score	0.941	0.824	0.769
Accuracy	0.961	0.913	0.833

$\text{FP} + \text{FN}$); $\text{recall} = \text{TP} / (\text{TP} + \text{FN})$; $\text{precision} = \text{TP} / (\text{TP} + \text{FP})$; and $\text{F1 score} = 2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall})$.

Results

A total of 1,000 panoramic radiographs were included in this study. The 2 raters (N.A. and M.M.) demonstrated very good IRR values of 0.93, 0.74, 0.89, and 0.87 for artifacts, positioning, coverage area, and contrast criteria, respectively.

Artifacts

The accuracy of the model on the validation dataset was 87.2%. Of the 312 images included in the validation dataset, 122 of 143 images with an artifact (class 0) and 150

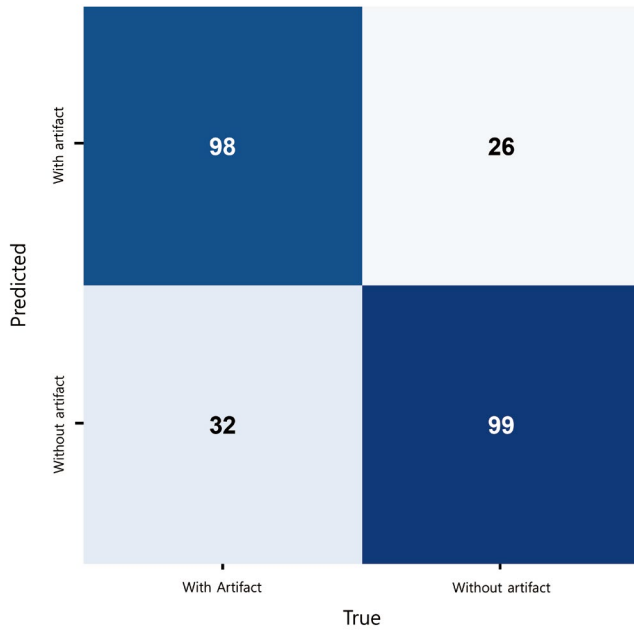


Fig. 8. Confusion matrix for patient positioning classification on panoramic radiographs.

Table 4. Model performance on classifying panoramic images according to patient positioning

Metrics	Class 0 (poor patient positioning)	Class 1 (good patient positioning)
Precision	0.643	0.778
Recall	0.818	0.583
F1 score	0.720	0.667
Accuracy	0.769	0.692

of 169 images without artifacts (class 1) were correctly classified using the proposed model (Fig. 6). The model performed better in determining the presence of an artifact in panoramic images.

Table 2 details the model performance according to this parameter on a set of new test images.

Coverage area

The highest accuracy for classifying images based on coverage area was 74.1% for the validation dataset. In this dataset, comprising 370 images, the proposed model correctly classified 101 of 135 images with poor coverage area (class 0), 66 of 110 images with moderate coverage area (class 1), and 107 of 125 images with good coverage area (class 2) (Fig. 7).

The model more accurately classified images with good and poor coverage areas than those with moderate cov-

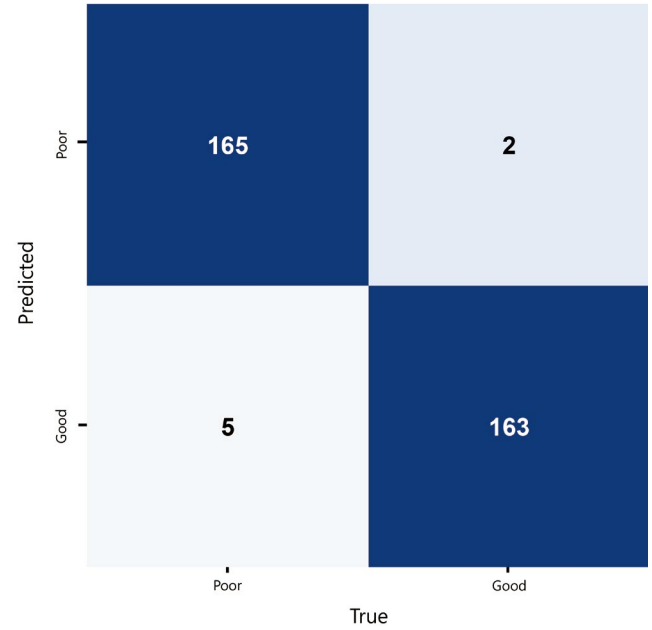


Fig. 9. Confusion matrix for contrast/density classification on panoramic radiographs.

Table 5. Model performance on classifying panoramic images according to contrast/density

Metrics	Class 0 (poor contrast/density)	Class 1 (good contrast/density)
Precision	0.9	0.667
Recall	0.692	0.889
F1 score	0.782	0.765
Accuracy	0.833	0.846

erage areas. However, on a set of new test images, the model performed slightly better in classifying images with poor and moderate coverage areas compared to those with good coverage areas (Table 3). Overall, the classification model for coverage area outperformed the other classification models in terms of accuracy on a new set of images.

Patient positioning

The model used for patient positioning achieved an accuracy of 77.3% on the validation dataset. Specifically, 98 of 130 panoramic images with poor patient positioning and 99 of 125 images with good patient positioning were correctly classified using the proposed model (Fig. 8). Thus, the performance in classifying the validation dataset into 2 groups based on patient positioning was approximately equal.

On a set of new test images, as depicted in Table 4, poor patient positioning was diagnosed more accurately than

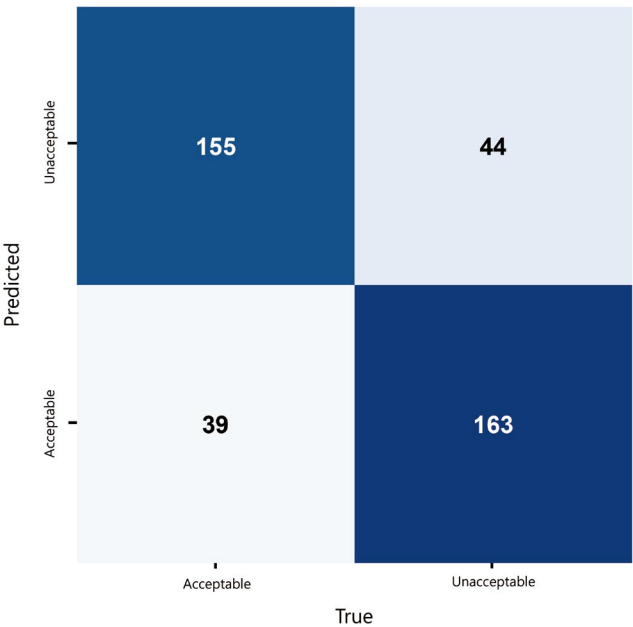


Fig. 10. Confusion matrix for overall quality classification on panoramic radiographs.

Table 6. Model performance on classifying panoramic images according to overall quality

Metrics	Class 0 (clinically unacceptable)	Class 1 (clinically acceptable)
Precision	0.844	0.788
Recall	0.776	0.854
F1 score	0.809	0.820
Accuracy	0.814	0.814

good patient positioning (accuracy of 0.77 vs. 0.69).

Contrast and density

The model used for classifying images based on contrast and density outperformed the other classification models on the validation dataset, achieving an accuracy of 97.9%. Of the 335 images included in the validation dataset, 165 of 170 images with poor contrast/density (class 0) and 163 of 165 images with good contrast/density (class 1) were correctly classified by the proposed model (Fig. 9).

On a set of new test images, as depicted in Table 5, the model demonstrated a balanced classification performance for images with poor and good contrast/density.

Overall quality of the radiographs

Regarding overall quality, the model achieved an accuracy of 79.3% on the validation dataset. In this dataset, the

proposed model correctly classified 155 of 194 images as clinically unacceptable (class 0) and 163 of 207 images as clinically acceptable (class 1) (Fig. 10).

Table 6 details the model’s performance on a set of new test images in classifying panoramic images based on the need to retake the radiograph for proper diagnosis. The model exhibited comparable performance in classifying images of high (clinically acceptable) and poor (clinically not acceptable) quality.

Discussion

The YOLOv8 models in the present study demonstrated strong performance in assessing various aspects of image quality, achieving classification accuracies of 87.2% for artifact detection, 74.1% for coverage area, 77.3% for patient positioning, 97.9% for contrast and density, and 79.3% for overall image quality. When determining whether an image was clinically acceptable, the model displayed an average accuracy of 81.4%, highlighting its potential for real-world application in automated dental image evaluation. In an era marked by rapid technological progress and advances in machine learning, the integration of AI has emerged as a key development across medicine and dentistry. Traditional IQA relies on subjective expert evaluations, which are time-consuming, prone to variability, and require considerable expertise. Although previous DL approaches have attempted to address this challenge, the current study is the first to employ YOLOv8, an advanced real-time object detection and classification algorithm, in panoramic radiograph quality assessment. Unlike conventional methods such as rule-based algorithms and handcrafted feature extraction, YOLOv8 leverages end-to-end learning to automatically extract relevant quality features from radiographs without the need for manually defined criteria.³³

Utilizing machine learning methods based on DL has led to notable progress in various medical imaging diagnoses and quality assessments. These models have demonstrated performance comparable to that of clinical experts in disease classification tasks.³⁴ At the core of developing these advanced models is the careful use of extensive, high-quality datasets. This approach enables the models to discern intricate patterns, recognize nuanced features, and ultimately improve performance.²¹

Quality assessment of dental radiographs is an integral component of the standard of care. Clinicians must ensure the accuracy of diagnostic information for precise clinical assessment, prioritize timely access to this information for effective diagnosis and intervention, and minimize radia-

tion exposure during radiographic examinations.³⁵

Recognizing the importance of high-quality images for training robust machine learning algorithms, Delamare et al.²¹ explored methods for addressing errors in panoramic radiographs used for training such models. Their systematic literature review categorized methods for managing low-quality radiographs into 3 groups. Some studies acknowledged the impact of low-quality input on model performance but took no measures to address it. A subset of studies incorporated quality assessment into their inclusion/exclusion criteria, thus excluding low-quality images from the input dataset. Another group employed image enhancement techniques to improve the quality of the radiographs. The review categorized errors such as distortion, artificial noise, and blur, as well as issues related to brightness, resolution, contrast, artifacts, and mispositioning, as factors potentially influencing image quality. The present study effectively incorporated nearly all criteria outlined in this systematic review, and the results and final classification demonstrate noteworthy accuracy and reliability.

The present study underscores the potential and effectiveness of employing a DL CNN YOLOv8 classification model for objective IQA. YOLOv8 is the latest version of the YOLO algorithm series.³⁶ Compared to previous CNN-based models such as VGG16, ResNet, and DenseNet, YOLOv8 offers a single-stage approach optimized for both speed and accuracy. Unlike multi-stage architectures such as Faster R-CNN, which require region proposal generation and multi-step processing, YOLOv8 performs a real-time assessment of panoramic radiographs without compromising classification performance.³⁷ In this study, the YOLOv8 architecture was used to categorize panoramic radiographs based on artifacts, coverage area, patient positioning, contrast/density, and, importantly, clinical acceptability. The model accuracy in detecting various errors affecting radiograph quality ranged from 74.1% for coverage to 97.9% for contrast and density, with an accuracy of 79.3% for overall radiograph quality and acceptability. Particularly strong performance was observed in detecting contrast and density issues, suggesting the ability to effectively extract features such as bright spots or low-contrast regions from radiographs. The lower accuracy in coverage area classification likely stems from the subjective nature of the assessment and the absence of clear boundaries between moderate, poor, and good coverage. Unlike contrast and density, which have distinct pixel intensity levels, coverage assessment relies on spatial distribution and anatomical landmarks that can vary with patient positioning and operator technique. Differences in collimation and over-

lapping structures further complicate classification. Future improvements could involve incorporating contextual cues, such as bounding box annotations or segmentation, to increase accuracy. Previous studies have also demonstrated the capability of DL models to identify areas of high contrast.³⁸

Additionally, the capacity of YOLOv8 to process full images—rather than segmenting patches, as is common with traditional CNN architectures—provides a more comprehensive IQA. This feature allows for improved detection of global radiographic errors, such as positioning and coverage issues, which may be overlooked by models that rely on localized patch-based analysis.³⁹

The proposed model surpassed existing methods, achieving precision and recall rates exceeding 90% after post-processing for tooth detection and numbering, as well as over 60% precision and recall in segmentation tasks. The YOLO series was also employed for treatment planning with dental panoramic radiographs, yielding noteworthy outcomes. The model achieved a precision of 100%, an F1 score of 76%, a recall of 96%, and an impressive mean Average Precision of 81.9%.

Despite these promising results, several limitations must be acknowledged. First, this study is limited by the inherent variability in radiographic image quality, which may affect model generalizability across clinical settings. Second, although the dataset was diverse, it does not fully represent all possible patient demographics and imaging conditions, potentially limiting the model's adaptability to new populations. Additionally, while YOLOv8 performs well as a single-stage classifier, its reliance on image-level labels rather than pixel-wise annotations may limit its ability to capture fine-grained variations in radiographic quality. Lastly, real-time deployment in clinical workflows could require further validation to ensure seamless integration and minimize disruptions to diagnostic efficiency.

A key limitation of this study was the difficulty in obtaining images with obvious quality errors across all 4 criteria for IQA. This challenge was addressed by applying augmentation methods to increase the number of images with poor quality in all 4 criteria, making them nearly comparable to the images of good quality. Additionally, the variability in the number of images across different categories was a limitation. Due to the limited availability of images with poor scores for each criterion, the authors were compelled to include different numbers of images in each group.

Given the critical role of large datasets in training machine learning models, previous studies have highlighted the need for machine learning tools for automatic image

classification.³⁸ The capability of YOLOv8 to classify radiographs in real time offers a considerable advantage for clinical integration, reducing radiographic retakes and minimizing errors caused by poor image quality. The future development of this model holds significant potential for implementation in both clinical practices and dental schools, as it facilitates objective radiograph classification and the identification of common acquisition issues that compromise radiographic imaging quality in dentistry.

Future research should prioritize increasing dataset diversity to increase the model's robustness in different patient demographics and imaging conditions. By incorporating a wider range of data, the model's performance can be ensured across diverse clinical scenarios. Another promising direction for exploration is multi-task learning, in which the model assesses multiple quality criteria simultaneously. This approach could improve accuracy by enabling the model to recognize shared features across tasks. Expanding the model's application to other imaging modalities, such as periapical and cone-beam computed tomography scans, could further demonstrate its versatility in broader diagnostic settings. Additionally, real-world validation studies are essential to determine how well the model integrates into clinical workflows, ensuring that it provides meaningful and practical support for dental professionals.

Despite these benefits, integrating AI into clinical and educational settings presents challenges. In clinical practice, seamless integration with existing systems, such as electronic health records, is necessary to provide real-time feedback without disrupting workflow. Regular validation with diverse datasets will be key to maintaining accuracy and reliability across different populations and imaging conditions.

In dental education, incorporating YOLOv8 into radiology training could involve AI-assisted modules that provide instant feedback on image quality. However, faculty and students may need guidance in interpreting AI-generated insights. Acceptance of AI tools in education and practice may also require thoughtful implementation; offering clear explanations through visual aids or confidence scores could help build trust. Understanding user experience and addressing concerns about interpretability will be critical for successful real-world adoption.

Furthermore, although CNN-based approaches such as ResNet and VGG have already been used for medical image classification,⁴⁰ future comparative studies could explore how transformer-based models, such as Vision Transformers,⁴¹ perform against YOLOv8 in terms of IQA efficiency and accuracy. Integrating this model into the

curriculum can provide students with hands-on experience using advanced diagnostic tools, assisting in their learning and practical skill development. This is particularly crucial for training students and technicians, as it aims to prevent recurring errors and ultimately reduce radiographic retakes, which are estimated to affect up to 22% of images.^{42,43}

When students are exposed to these technologies, they can better understand the importance of radiographic quality and the impact of errors, thus becoming more competent and confident practitioners. Such advancements could improve the quality of care by reducing the patient dose associated with radiographic retakes, supporting the training of dental staff, and streamlining clinic workflow. Furthermore, the ability to objectively assess and correct radiographic quality issues during their education will prepare students to implement these practices in their future careers, promoting a culture of continuous improvement and excellence in dental radiography.

By employing 4 criteria to evaluate the quality of panoramic radiographs and training a YOLOv8 DL model for objective quality assessment, the results of this study demonstrate promising outcomes that affirm the feasibility of the model in classifying radiographs. The integration of such advanced AI-driven tools in dental education can augment the training of future dental professionals by providing them with real-time feedback on radiographic quality. This would not only improve their diagnostic skills but also instill a deeper understanding of the importance of high-quality imaging in clinical practice. However, this study has certain limitations, including dataset variability and the challenges of generalizing the model across diverse imaging conditions. Further research is needed to expand dataset diversity, refine classification accuracy, and validate the model's performance in real-world clinical settings. Nevertheless, exposure to cutting-edge technology during their education prepares students to adeptly incorporate these innovations into their future practices, ultimately elevating the standard of patient care.

Conflicts of Interest: None

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